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Development of a UAV Surveillance System Using Deep Learning and Thermal Imaging for High-Risk Environments

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Abstract

Unmanned Aerial Vehicles (UAVs) are becoming integral components of Industry 4.0 security architectures, offering autonomous, data-driven monitoring of industrial facilities and other high-risk environments. This work presents the development of a modular UAV-based surveillance system that integrates deep-learning-powered object detection, real-time thermal imaging, and semi-autonomous navigation for enhanced situational awareness. Using a modified Tello drone equipped with a Raspberry Pi 4, ESP wireless module, GPS, and MLX90614 thermal sensor, the system performs real-time visual and infrared analysis to detect, classify, and track humans with high accuracy. Object detection is achieved through a custom-trained YOLOv8 model, incorporating a public dataset to ensure robustness under varied lighting and environmental conditions. A Python-based graphical interface provides integrated manual control, autonomous waypoint navigation, and live data overlays, enabling seamless operator interaction. Structural and aerodynamic optimisations using SOLIDWORKS ensure flight stability despite additional payloads. Field tests in dynamic outdoor and indoor industrial scenarios demonstrate strong performance: mean Average Precision (mAP50-95) of 87%, precision of 95%, and recall of 90%. Unlike prior works that address either visual or thermal detection separately, this system delivers a compact, low-cost, and scalable surveillance platform tailored for smart manufacturing and security-sensitive operations, supporting predictive security monitoring, PPE compliance verification, and automated perimeter inspection in Industry 4.0 environments.

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Keywords: UAV surveillance, YOLOv8 object detection, Thermal imaging, Autonomous navigation, Industry 4.0 security, Smart manufacturing monitoring.

1. Introduction

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have emerged as a transformative technology in surveillance, security, and industrial monitoring, offering rapid deployment, wide-area coverage, and high-resolution data acquisition. In the context of Industry 4.0, UAVs play an increasingly important role as mobile sensing platforms

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within cyber-physical systems, providing real-time situational awareness and integrating seamlessly with industrial IoT networks. Their ability to autonomously navigate, capture multi-modal imagery, and transmit data in real time makes them valuable for addressing security challenges such as perimeter intrusions, theft prevention, and safety compliance monitoring in smart manufacturing environments [1,2]. Despite these advantages, conventional UAV surveillance systems face limitations, including reliance on manual operation, reduced effectiveness in low-visibility conditions, and insufficient integration with predictive industrial security frameworks [3]. This work addresses these gaps through the development of a modular, AI-enhanced UAV surveillance system that combines deep learning-based object detection, thermal imaging, and semi-autonomous navigation. The platform is built on a modified Tello drone equipped with a Raspberry Pi 4, ESP wireless module, GPS, and MLX90614 thermal sensor, controlled via a custom Python-based graphical user interface (GUI) [4]. The novelty of this work lies in its integration of YOLOv8-powered object detection with real-time thermal imaging on a compact, low-cost UAV platform, coupled with semi-autonomous waypoint navigation and a modular design that supports sensor interchangeability [5, 6]. The system is optimized for smart manufacturing environments, where it can detect and track individuals, verify PPE compliance, and monitor high-value industrial assets both indoors and outdoors, even under challenging lighting conditions. Structural and aerodynamic enhancements ensure stability despite additional payloads, and the modular architecture allows for future upgrades such as LiDAR integration and fully autonomous navigation [7, 8].

This paper contributes to the field by (1) demonstrating a cost-effective UAV surveillance system tailored for Industry 4.0 security requirements, (2) detailing the integration of multi-modal sensing with AI-driven analytics for real-time decision support, and (3) validating performance through field tests in both daylight and low-light scenarios [9]. The proposed system advances UAV-assisted surveillance from basic visual monitoring to predictive, IoT-connected industrial security, supporting proactive incident prevention and operational continuity in next-generation manufacturing facilities.

2. Review of Literature

2.1 Background and Evolution of UAVs

Unmanned Aerial Vehicles (UAVs) have evolved from their military origins to diverse civilian applications, including surveillance, disaster management, and industrial monitoring. Early UAVs were primarily deployed for reconnaissance missions, but advancements in automation, artificial intelligence, and sensor technology have expanded their capabilities [10]. Today, UAVs play a vital role in security, with applications in law enforcement, border control, and intelligence gathering [11]. However, challenges such as battery life, flight endurance, and environmental adaptability remain key concerns [12]. One of the most critical aspects of UAV development is the integration of real-time data processing with deep learning algorithms for enhanced object recognition. Recent studies have explored AI-powered UAVs for intelligent surveillance, combining high-resolution imaging with advanced tracking models to improve accuracy in identifying and following targets [13].

2.2 UAVs in Surveillance and Security

(a) Object Detection and Tracking - Modern surveillance UAVs employ sophisticated object detection models such as YOLO (You Only Look Once) to identify and classify objects in real-time. These models leverage convolutional neural networks (CNNs) to enhance detection accuracy and speed [14]. Research has demonstrated that deep learning-based approaches significantly outperform traditional image processing methods in tracking moving targets [15]. YOLOv3 and its subsequent improvements, such as YOLOv8, have been widely adopted in UAV-based security applications due to their efficiency in processing aerial imagery [16]. The integration of thermal imaging sensors has further enhanced UAV surveillance, allowing for target detection in low-visibility conditions. Studies have shown that thermal-assisted UAVs improve situational awareness during night operations and adverse weather conditions [17]. Additionally, multi-camera setups on UAVs facilitate improved depth perception, leading to more reliable tracking performance [18].

(b) *UAV-Assisted Crime Prevention and Border Security* - UAVs have proven valuable in counteracting criminal activities such as smuggling, poaching, and illegal border crossings. The deployment of drones for real-time

monitoring of high-risk zones has been instrumental in reducing response times for law enforcement agencies [19]. The Armed Conflict Location and Event Data Project (ACLED) reports that UAV surveillance has been increasingly utilized in addressing security threats such as kidnapping and armed conflicts in regions like Nigeria [20]. Recent studies highlight the effectiveness of UAVs in urban security, where drones assist in crowd monitoring, anomaly detection, and emergency response coordination [21]. Law enforcement agencies worldwide are exploring AI-powered UAVs for predictive policing, using machine learning models to analyze behavioral patterns and anticipate criminal activities [22].

2.3 Advances in UAV Technology

(a) Deep Learning for UAV Navigation - Deep learning algorithms have significantly improved UAV autonomy, enabling real-time decision-making in complex environments. Reinforcement learning approaches have been applied to UAV navigation, allowing drones to optimize flight paths while avoiding obstacles [23]. Researchers have also developed hybrid AI models that combine deep learning with traditional control algorithms to enhance UAV stability and responsiveness [24]. In addition to autonomous navigation, AI-driven UAVs are now capable of real-time anomaly detection, allowing for the identification of unusual events or movements. These advancements have critical applications in disaster management, infrastructure inspection, and environmental monitoring [25].

(b) Communication and Data Processing in UAV Networks - The integration of UAVs with Internet of Things (IoT) systems has enhanced their real-time data processing capabilities. IoT-enabled UAVs can relay live data streams to cloud-based platforms for instant analysis, improving decision-making in surveillance operations. Furthermore, advancements in 5G technology have enabled low-latency communication between UAVs and ground control stations, enhancing real-time control and coordination. Efficient energy management remains a significant challenge in UAV technology. Researchers are exploring wireless charging solutions and solar-powered UAVs to extend operational endurance, addressing a key limitation in long-duration surveillance missions.

2.4 Future Trends in UAV-Assisted Surveillance

Emerging trends in UAV technology focus on enhancing automation, multi-agent coordination, and swarm intelligence. The deployment of UAV swarms, where multiple drones operate collaboratively using AI-based coordination, is expected to revolutionize large-scale surveillance and search-and-rescue operations. Additionally, the integration of LiDAR technology with UAV systems is anticipated to improve 3D mapping capabilities, providing more accurate terrain analysis and obstacle detection. Despite these advancements, ethical concerns regarding UAV surveillance persist, particularly in relation to privacy and data security. Researchers are actively investigating regulatory frameworks to ensure responsible deployment of UAV technology while balancing security and civil liberties.

3. Materials and Method

3.1 Design Rationale

This work demonstrates not only a practical advancement in UAV-based security surveillance but also a strategic enabler for Industry 4.0, where integrated sensing, AI-driven analytics, and autonomous mobility converge to protect and optimize smart manufacturing environments. The system was designed with three guiding objectives: (1) to develop a lightweight, modular UAV platform capable of integrating multi-modal sensors for robust performance in varied lighting conditions; (2) to enable real-time, AI-driven object detection and tracking for both security and industrial safety applications; and (3) to ensure compatibility with Industry 4.0 environments through IoT-ready data processing and scalable deployment. The Tello drone, shown in Figure 1, was selected for its programmability, cost-effectiveness, and suitability for indoor/outdoor operations. YOLOv8 was chosen as the detection framework for its combination of high accuracy, fast inference, and compatibility with embedded hardware. Thermal imaging was incorporated to address the limitations of visual-spectrum detection under low-light or obscured conditions. The UAV system integrates four core modules:

i. UAV Platform – A modified Tello drone with additional onboard computing and sensing capabilities.
ii. Object Detection and Thermal Imaging – YOLOv8 for object recognition, combined with an MLX90614 thermal sensor for heat signature analysis.

iii. Control and Communication – A custom Python-based GUI enabling manual and semi-autonomous operation with live telemetry and video feed.
iv. Data Processing and Visualization – Real-time overlay of detection results on the video stream, with storage for post-mission analysis.

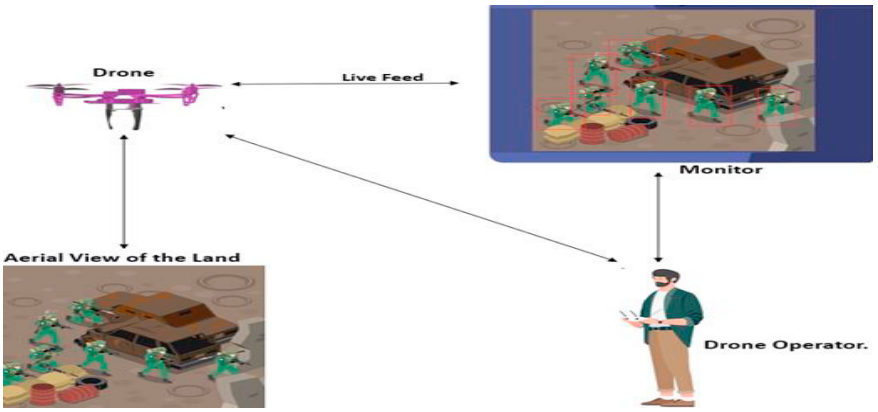


Figure 1. System Architecture

3.2 Hardware Selection and Integration
The hardware components were selected based on their weight, efficiency, and compatibility with the UAV system.

3.2.1 Tello Drone
The Tello drone was chosen due to its lightweight design, ease of programmability, and built-in camera [10]. Compared to building a drone from scratch, the Tello provided a cost-effective alternative with adequate processing power to support object detection and thermal imaging. Additional enhancements included integrating a Raspberry Pi 4 for onboard computing, an ESP module for wireless communication, and a thermal sensor for environmental analysis.

3.2.2 Object Detection and AI Model Implementation
A YOLOv8 model was trained on a Ultralytics dataset consisting of Visdrone images 2163 using one object category person, captured in daylight, and low-light. Images were annotated using Pascal Vic-style bounding boxes for humans and relevant objects. The dataset was split into 70% training, 20% testing, and 10% validation. Augmentation techniques (Mosaic, random flip, hue-saturation-value HSV adjustments) improved generalization. Training was performed on an Google Collab T4 model (batch size: 8GB, learning rate: 0.01, 100 epochs), achieving convergence with Optimizer- AdamW. A YOLOv8 deep learning model was trained for object detection. The model was implemented using OpenCV and PyTorch, enabling real-time inference and tracking of objects within the drone's field of view [14]. The detection workflow followed these steps:

Image Acquisition – The drone's camera captures frames, which are preprocessed before analysis.
Object Classification and Bounding Box Prediction – YOLOv8 predicts object locations and labels with confidence scores.
Post-Processing and Data Visualization – Results are displayed in the GUI and stored for further evaluation.
The model's performance was validated using Mean Average Precision (mAP) and Intersection over Union (IoU) metrics [15]. These metrics ensured the system could accurately detect and track human movement, essential for security applications.

3.2.3 Heat Map Sensor Integration
A thermal sensor was incorporated to detect heat signatures in low-visibility environments. This was particularly useful for identifying individuals at night or in densely vegetated areas [17]. The sensor was interfaced with the ESP module, which transmitted data to the Raspberry Pi 4 for processing. To improve accuracy, the system applied

temperature calibration techniques, ensuring that false positives were minimized. Thermal data was overlaid onto the video feed, providing a composite view of both optical and infrared information.

3.3 Software Development and System Control

(a) Python-Based GUI for Drone Control - A custom Graphical User Interface (GUI) was developed using Pygame, replacing the default Tello application. This interface enabled Real-time video streaming and telemetry monitoring; Object detection and tracking overlay and Manual and autonomous flight controls. By integrating the GUI with the Tello SDK, commands such as takeoff, landing, and directional movement were mapped to intuitive button inputs [16].

(b) Navigation and Object Tracking - The drone was programmed to operate in two modes which include the manual and autonomous mode. For manual mode, the user controls the drone via the GUI. For the autonomous mode, the drone follows predefined waypoints and adjusts its path based on detected objects. Waypoint execution was implemented using a predefined navigation sequence without dynamic path re-planning. Target tracking was performed through a camera-based detection module, which maintained lock while the target remained in frame. The absence of adaptive path adjustment limited the system's ability to sustain tracking during occlusions or abrupt target movements. Integrating reinforcement learning-based path optimization and sensor fusion for obstacle detection could enhance collision avoidance and overall flight stability in future iterations. [23].

3.4 Simulation and Testing

(a) Aerodynamic and Structural Testing - To ensure flight stability despite additional hardware, the drone's aerodynamics were tested using SolidWorks simulations. The lift-to-weight ratio and drag coefficients were analyzed to optimize payload distribution and minimize performance degradation [24].

(b) Object Detection Model Evaluation - The YOLOv8 model's effectiveness was evaluated using:
IoU – Measuring the overlap between predicted and actual bounding boxes.
mAP – Assessing detection precision across different confidence thresholds.
Testing was conducted in various environments to analyze performance under variable lighting, occlusion, and movement conditions [14]. Performance was validated using IoU, mAP, and loss reduction metrics as presented in Table 1.

Table 1. Object Detection Model Evaluation

Metric	Description	Result
IoU	Overlap between predicted and actual bounding boxes	82%
mAP50-95	Accuracy across different IoU thresholds	87%
Precision	Correctly detected objects vs. false positives	95%
Recall	Captured all relevant objects	90%

(c) Field Testing and Performance Metrics - The drone system was tested in real-life case scenarios, focusing on:
Latency in video processing and transmission; Accuracy of thermal imaging in human detection; Effectiveness of autonomous flight in dynamic environments. Performance data was logged and analyzed to identify areas for improvement.

3.5 Limitations and Future Enhancements

While the proposed system demonstrates significant improvements in UAV-assisted surveillance, certain limitations remain which include the battery life, environmental factors and integration with LiDAR. The information is further expanded thus:

i. Battery life constraints – The added hardware components reduce flight endurance. Future work will explore solar-powered UAVs and battery optimization strategies.
ii. Environmental factors – Wind resistance and extreme temperatures can impact UAV stability. Enhancements in adaptive flight control and weather-resistant components are planned.
iii. Integration with LiDAR – Future iterations of the system will incorporate LiDAR for enhanced 3D mapping and obstacle detection.

4. Results and Experiment

This section presents the results obtained after conducting pre-tests and post-tests to evaluate the UAV system's functionality. The tests focused on autonomous flight, object detection, human tracking, and thermal imaging

4.1 Pre-Test Evaluations

Before full deployment, several preliminary tests were conducted to validate the system's flight stability, object detection accuracy, and sensor integration.

4.1.1 Autonomous Flight Performance

To test the drone's autonomous navigation, predefined GPS coordinates were set, allowing it to move from Point A to Point B without manual control (see Figure 2).



Figure 2. Autonomous Flight of Drone

The system successfully executed precise, unassisted movements, confirming the accuracy of its GPS-based navigation. The parameter for the autonomous flight is presented in Table 2.

Table 2. Parameters for Autonomous Flight

Parameter	Description	Example Value	Units	Source	Usage
Longitude	East-west coordinate	5.735	°	GPS Module	Determines location
Latitude	North-south coordinate	5.519	°	GPS Module	Determines location
Altitude	Height above sea level	50	m	Barometer	Maintains operational height
Speed	Drone's velocity	5	m/s	IMU/GPS	Ensures efficient movement
Orientation	Pitch, roll, and yaw	Pitch: 2°, Roll: 0°, Yaw: -15°	°	Gyroscope	Stabilization control
Obstacle Detection	Detects nearby objects	Yes	Boolean	Proximity Sensor	Collision avoidance

4.1.2 Object Detection with YOLOv8

The YOLOv8 model was employed for real-time human detection, enabling the drone to track individuals and capture automated snapshots. An example of a detected object is presented in Figure 3.



Figure 3. Object Detection

The detection system effectively identified and tracked humans, highlighting its potential for surveillance applications.

Table 3. Parameters for Object Detection

Parameter	Description	Value	Units	Source	Usage
Bounding Box	Object location in image	(120, 200, 300, 400)	Pixels	ML Model	Localizes objects
Detection Class	Type of object detected	Human	Text	ML Model	Categorizes targets
Confidence Score	Detection accuracy	95	%	ML Model	Ensures reliability
Segmented Area	Object size in image	2500	Pixels²	Segmentation Model	Measures detected object

4.1.3 Human Counting and Tracking

To enhance surveillance, the system was programmed to assign unique IDs to detect humans and track their movement across a predefined boundary. The system successfully counted individuals crossing a defined region, proving its usefulness in crowd monitoring as shown in Figure 4. Table 5 shows the parameters of human tracking.

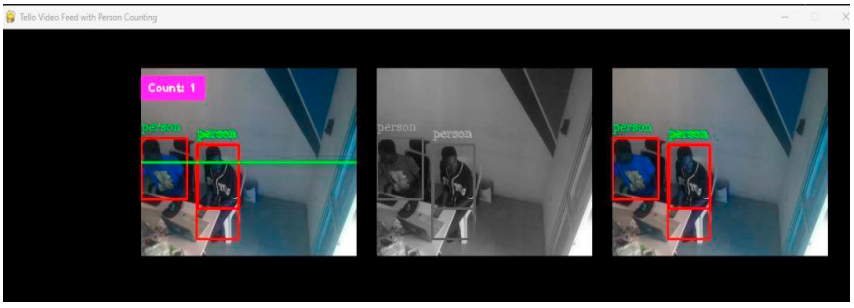


Figure 4: Image showing Object Detection and Human Count

Table 4. Parameters for Human Tracking

Parameter	Description	Value	Units	Source	Usage
Object ID	Unique identifier	1	Numeric	Tracking Algorithm	Differentiates people
Current Position	Object coordinates	(150, 300)	Pixels	Tracking Algorithm	Updates object location
Movement Path	Tracked trajectory	[(120, 200), (130, 220)]	Pixels	Tracking Algorithm	Records object motion

4.2 Post-Test Evaluations

4.2.1. Precision, Recall, and Detection Accuracy

The object detection model's accuracy was evaluated using precision, recall, and Mean Average Precision (mAP). The precision measures false positive rate, ensuring correct object classification. Recall evaluates how many relevant objects were detected within a specified period and the IoU (Intersection over Union) measures bounding box accuracy. Based on the test result, Table 4 presents the metrics value for IoU, precision and recall. The graphical plot is presented in Figure 5.

Table 5. Metrics Result

Metric	Description	Result
IoU	Bounding box overlap accuracy	82%
mAP50-95	Mean Average Precision across IoU thresholds	87%
Precision	Avoidance of false positives	95%
Recall	Capturing all relevant objects	90%

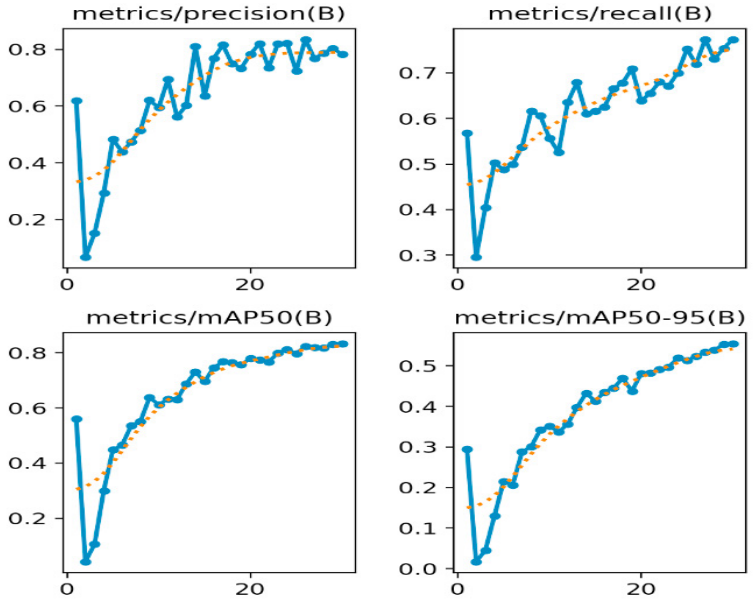


Figure 5. Detection Model Performance Metrics

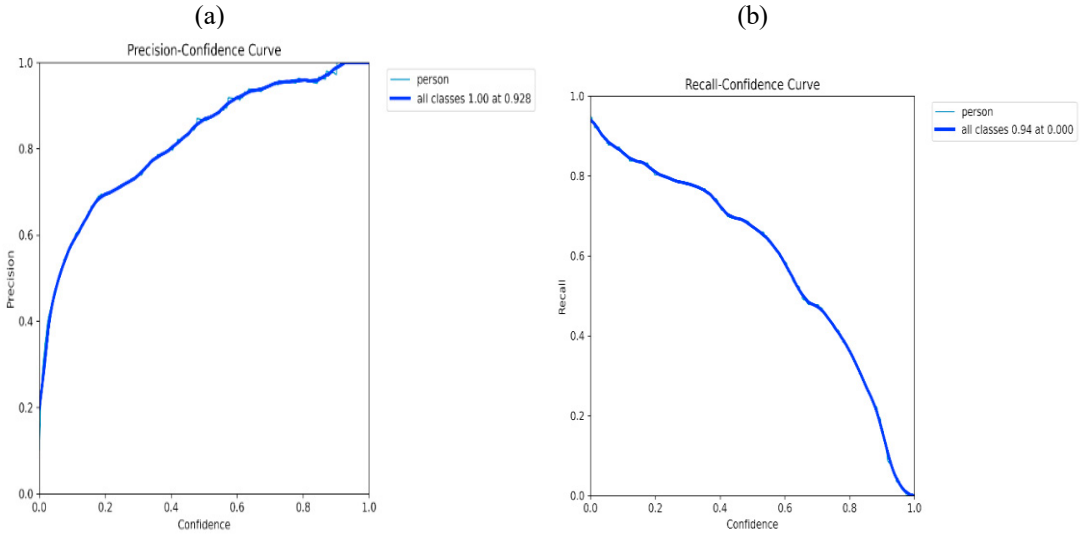


Figure 6 (a) Precision-Confidence Curve (b) Recall-Confidence Curve

The results indicate that the system maintains high accuracy in detecting humans in real-life situations. Figure 6 (a) represents the precision confidence curve after testing while Figure 6(b) shows the recall confidence call

4.2.2 Loss Reduction Analysis

The model's classification loss, box loss, and focal loss steadily decreased over training cycles, improving object detection accuracy. The Box Loss represent the accuracy of bounding box placement. The classification loss represent the model's ability to differentiate objects. The distribution focal loss has to do with fine-tuning bounding box precision. Figure 7 shows the train/loss and val/loss graphs.

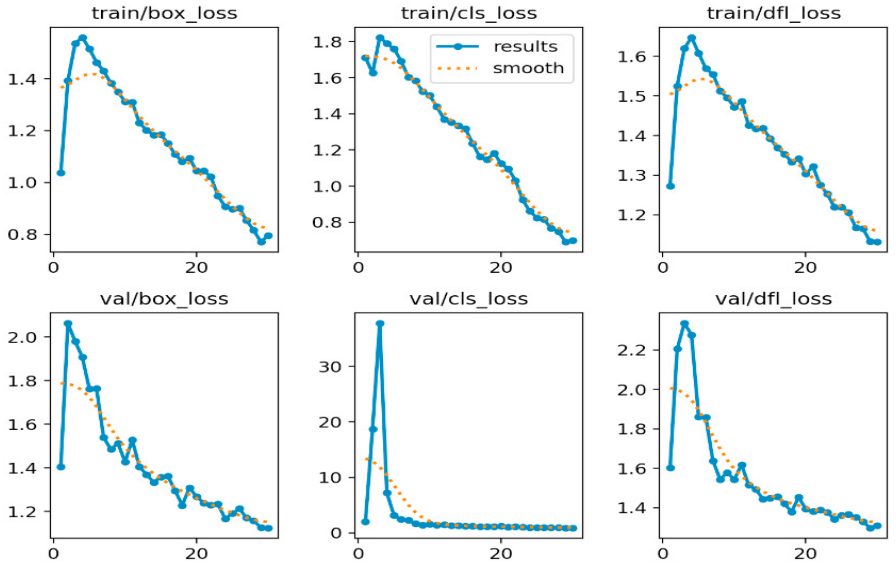


Figure 7: Train/Loss and Val/Loss Graphs

The consistent loss reduction confirms that the model efficiently generalizes to unseen images, reducing false detections.

4.2.3 Heat Map Visualization for Thermal Detection

The MLX90614 thermal camera was used to generate real-time heat maps, improving object visibility in low-light environments. The heat maps effectively highlight human presence based on thermal signatures, demonstrating the system's suitability for nighttime surveillance. Image generated in real-time heat map visualization is presented in Figure 8.

4.3 Challenges and Future Improvements

Limitations: Although the UAV demonstrated strong performance, several limitations were identified. First, battery life constraints limit operational time, suggesting future exploration of solar-powered UAVs. Second, environmental sensitivity was observed, as strong winds affected flight stability, highlighting the need for adaptive flight controls. Third, latency in data processing remains a challenge, with improvements in real-time video transmission speed required to enhance responsiveness.

Proposed Enhancements: To address these limitations, several improvements are proposed. LiDAR integration would enhance 3D mapping and obstacle detection. AI-based flight optimization using reinforcement learning could improve path planning and navigation. Finally, advanced power management strategies, including wireless charging and more efficient battery utilization, would extend mission endurance.

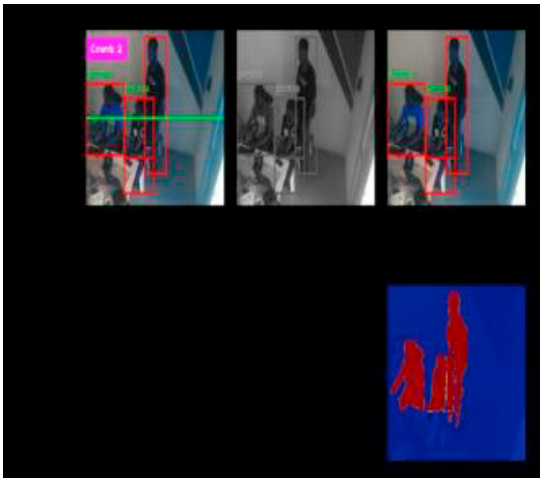


Figure 8. Real-Time Heat Map Visualization

5. Conclusion

This study has successfully demonstrated the feasibility and effectiveness of a modular UAV-assisted surveillance system that integrates YOLOv8-based object detection, real-time thermal imaging, and semi-autonomous navigation into a compact, low-cost platform. Through hardware enhancements, a custom Python-based GUI, and rigorous field testing, the system achieved a mean Average Precision (mAP50-95) of 87%, a precision of 95%, and a recall of 90%, maintaining high detection accuracy in both daylight and low-light conditions. Unlike existing solutions that focus on either visual or thermal detection independently, the proposed design delivers a multi-modal, real-time monitoring capability that is both scalable and adaptable to diverse environments. Structural and aerodynamic optimizations ensured stable flight despite the addition of extra payloads, while the modular architecture supports future integration of advanced sensors such as LiDAR. Beyond its application in public safety, this system addresses a critical gap in Industry 4.0 security by enabling automated perimeter inspections, PPE compliance monitoring, and predictive surveillance in smart manufacturing facilities. By combining affordability, portability, and advanced AI-driven analytics, the platform represents a practical step toward intelligent, connected security systems that protect people, assets, and operations in next-generation industrial environments.

Recommendations for Future Research

Future development of the system will focus on several key areas aimed at enhancing its capability, reliability, and industrial applicability. Operational endurance will be extended through the incorporation of solar-assisted charging systems and high-capacity lithium-ion batteries, reducing downtime between missions. Navigation will advance from semi-autonomous waypoint control to fully AI-driven mission planning with real-time path optimization, obstacle avoidance, and autonomous decision-making. The sensor suite will be upgraded with LiDAR and multi-sensor fusion to enable high-resolution 3D mapping, improved obstacle detection, and detailed industrial asset inspection. Intelligent identification will be achieved through the deployment of an Intelligent Lookup Model (ILM) connected to secure industrial or national databases via encrypted APIs for real-time verification of detected individuals. The detection models will be expanded to address industrial safety applications, including PPE compliance monitoring, fire hazard detection, and thermal diagnostics of critical equipment. Full integration with IoT networks and Manufacturing Execution Systems (MES) will enable automated logging, predictive analytics, and centralized operational control. Finally, robust cybersecurity protocols, including end-to-end encryption of all video and telemetry data, will be implemented to ensure compliance with international data protection and privacy standards.

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